Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/26/2006
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Project Title:	Test of Lorentz and CPT symmetry with an ultra-sensitive atomic magnetometer		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We propose to conduct a precision test of Lorentz and CPT symmetry using a self-compensated He-3 atomic magnetometer. A weak magnetic-like field that couples differently to K and He-3 spins would be a sign of Lorentz or CPT violation generated by quantum gravity effects. We recently demonstrated world-record magnetic field sensitivity in a spin-exchange relaxation free (SERF) K magnetometer as well as efficient self-compensation of the external magnetic fields in a He-3 co-magnetometer. In addition to precision tests of Lorentz and CPT symmetry we propose to further develop the K magnetometer for more general applications. 1) Improve long-term stability of He-3 co-magnetometer 2) Collect data to set a new limit on Lorentz and CPT violation 3) Develop the next generation of the co-magnetometer experiment 4) Develop practical applications of sensitive magnetometery techniques Lorentz and CPT symmetry form the foundation for Standard Model and field theory. Any violation of these symmetries would give		

	an umambigous sign of new physics, most likely due to quantum gravity effects. In addition, precision measurement techniques developed for this project can be used for other applications, such as detection of weak magnetic fields, detection of MRI and NMR, and gyroscope applications
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will improve our understanding of fundamental physics of the Universe, result in development of new precision measurement techniques and training of graduate students.
Task Progress:	During the last year we have made significant progress on four fronts. We finished a long data collection process for the Lorentz violation test, made additional improvements to the apparatus to increase sensitivity at low frequency, started building a new system that will allow rotation of the apparatus to increase the frequency of modulation for the expected Lorentz signal, and investigated the use of the co-magnetometer as an inertial rotation sensor. In the first half of the year we focused on collection and analysis of data with stationary apparatus. The main challenge for obtaining high sensitivity is to reduce low frequency noise. We have implemented elaborate feedback systems to control the temperature of various components of the experiment, the position and power of lasers, and other parameters. The data collection was fully automated and computer controlled. During data collection, magnetic field offsets and other imperfections were periodically measured and set to zero. Data were collected over a period of approximately 1 year, with some interruptions for improvements in the apparatus. In addition to Lorentz-violating signal, about 30 other signals were monitored. We performed a detailed correlation analysis, looking for various spurious effects with a diurnal variation. Using these data we were able to set a limit on Lorentz and CPT-violating interaction for neutrons equal in sensitivity to the best existing limit. The limit on proton coupling is a factor of 100 better than existing limits. We are continuing to improve the sensitivity of the stationary system. Recently we replaced external cavity lasers used for optical pumping and probing with DFB lasers that have better frequency stability. DFB lasers also can be continuously tuned over tens of GHz without mode hopes, allowing us to zero light shift effects. We also made new K-3He cells with higher 3He pressure. Having a higher pressure increases the equilibrium magnetization and reduces the response time of the system, allowing faster alignment procedure. Cells with higher pressure also have smaller optical thickness, resulting in more uniform optical pumping and allowing operation at higher alkali-metal density. We also made a number of improvements in the optical and mechanical arrangement of the apparatus, reducing the effects of optical interference. We plan to continue taking data with the improved system in the coming year. At the same time, we are building a new apparatus that can be rotated in the lab. This will improve the sensitivity by moving the expected signal away from 1/f noise. The apparatus will be improved in several ways. It will use an electrically-insulating ferrite magnetic shield to reduce magnetic noise from Johnson currents. All optics will operate in vacuum, to reduce the effects of air convection on laser beams. The apparatus will also be much more compact, which increases the magnetic shielding factor and improves mechanical stability. We have developed and tested various components of the system. Instead of using hot air flow, the cell is heated by electrical heaters wound with zero dipole moment and operating on AC current. The heaters use glass-isolated Constantan wire imbedded in BN2 plate. We verified that magnetic fields from this heating system are sufficiently suppressed so they do not disturb operation of the magnetometer. We also tested a new technique for measuring optical rotation using a photoelastic modulator instead of a Faraday rotator. The photoelastic modulator has a higher modulation frequency and better temperature stability. We showed that low-frequency noise is substantially reduced with this new system. We made a detailed magnetic, thermal and mechanical design for compact magnetic coils, shields, optical system and rotation system. Currently we are in the process of ordering various components for the new apparatus. As a side project, we investigated the application of the co-magnetometer for inertial rotation sensing. Rotation of the apparatus is equivalent to a pseudo-magnetic field that couples to spins not in proportion to their magnetic moment. We showed that the co-magnetometer makes a sensitive nuclear spin gyroscope and tested various aspects of its performance. We studied theoretically the response of the system to transient magnetic fields and rotations and found that the integral of the co-magnetometer signal is always proportional to the total rotation angle, independent of the rotation rate. Spin precession due to magnetic field is also effectively cancelled in this arrangement. We also investigated the possibility of using 21Ne instead of 3He in the co-magnetometer. Because 21Ne magnetic moment is smaller than 3He by a factor of 10, its sensitivity to anomalous Lorentz coupling and inertial rotation is increased by the same factor. Also, since it has a nuclear spin equal to 3/2, it is sensitive to additional Lorentz-violating terms that cannot be detected with spin-1/2 particles. We have demonstrated operation of 21Ne co-magnetometer and studied several of its properties, such as quadrupole spin relaxation of 21Ne. We found that the spin-exchange rate with K atoms is insufficient to obtain large 21Ne polarization and plan to switch to using Rb atoms. Currently we are constructing a dedicated apparatus for measurement of various spin-exchange and spin-relaxation rates between 21Ne and alkali metals.
Bibliography Type:	Description: (Last Updated: 09/18/2007)
Articles in Peer-reviewed Journals	Kornack TW, Ghosh RK, Romalis MV. "Nuclear spin gyroscope based on an atomic co-magnetometer." Phys Rev Lett. 2005 Dec 2;95(23):230801. PMID: 16384290 , Dec-2005